IT4010 Research Project – 2025 July

Topic Assessment Form

Project ID : 25-26J-130

1. Topic (12 words max)

Multimodal AI Tutoring System.

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Data Science (DS)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

Even though educational technology is advancing quickly, most intelligent tutoring systems (ITS) still react to student errors or requests without any strategic planning. This limits their ability to support personalized, ongoing learning paths. Additionally, traditional ITS often depend on fixed rule-based systems that lack the flexibility to adjust in real time to different student actions, academic levels, or cognitive states.

Recent developments in artificial intelligence (AI), such as computer vision (CV), large language models (LLMs), reinforcement learning (RL), and agentic planning, offer new opportunities for creating proactive, adaptive, and multimodal tutoring systems. However, the challenge remains to combine these technologies into a single, intelligent framework that personalizes learning experiences in real time.

There is a significant gap in research focused on creating an AI-driven tutor that can mimic the behaviors of a goal-oriented human tutor. This tutor should be able to set long-term learning goals, break them into step-by-step instructional plans, adjust strategies based on student performance and engagement, and provide personalized feedback in various ways.

Current tutoring systems do not adequately use advanced planning methods, such as Tree of Thought or ReAct frameworks, which can improve foresight and intentional instructional reasoning. Furthermore, computer vision is seldom applied to assess real-time student attention and emotion, while reinforcement learning is underutilized in live, feedback-driven learning environments.

This research seeks to address these gaps by proposing a multimodal AI tutoring system that combines four key components: an agentic AI core for planning, an RL engine for optimizing strategies, an LLM-based content generator for tailored delivery, and a CV module for monitoring engagement in real time. These technologies aim to create a tutor that not only reacts but also actively guides learning through thoughtful, emotionally aware decision-making.

Y. Chen, N. Ding, H.-T. Zheng, Z. Liu, M. Sun, and B. Zhou, "Empowering Private Tutoring by Chaining Large Language Models," *arXiv preprint arXiv:2309.08112*, 2023. [Online]. Available: https://arxiv.org/abs/2309.08112.

A. M. Ragab, E. F. T. El Shenawy, and N. A. Hikal, "Advancing Generative Intelligent Tutoring Systems with GPT-4: Design, Evaluation, and a Modular Framework for Future Learning Platforms," *Electronics*, vol. 13, no. 24, p. 4876, 2024. [Online]. Available:https://www.mdpi.com/2079-9292/13/24/4876.

Z. Jiang and M. Jiang, "Physics-STAR: Beyond Answers: LLM-Powered Tutoring System in Physics Education for Deep Learning and Precise Understanding," *arXiv preprint arXiv:2406.10934*, 2024. [Online]. Available: https://arxiv.org/abs/2406.10934.

S. Yin, C. Fu, S. Zhao, K. Li, X. Sun, T. Xu, and E. Chen, "A Survey on Multimodal Large Language Models," *arXiv preprint arXiv:2306.13549*, 2023. [Online]. Available: https://arxiv.org/abs/2306.13549.

Q. Ye, H. Xu, G. Xu, J. Ye, M. Yan, Y. Zhou, and F. Zheng, "mPLUG-Owl: Modularization Empowers Large Language Models with Multimodality," *arXiv preprint arXiv:2304.14178*, 2023. [Online]. Available: https://arxiv.org/abs/2304.14178.

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

The proposed solution is a Multimodal AI Tutoring System that combines agentic reasoning, adaptive learning, and real-time perception to deliver proactive and personalized instruction. It consists of four tightly integrated components:

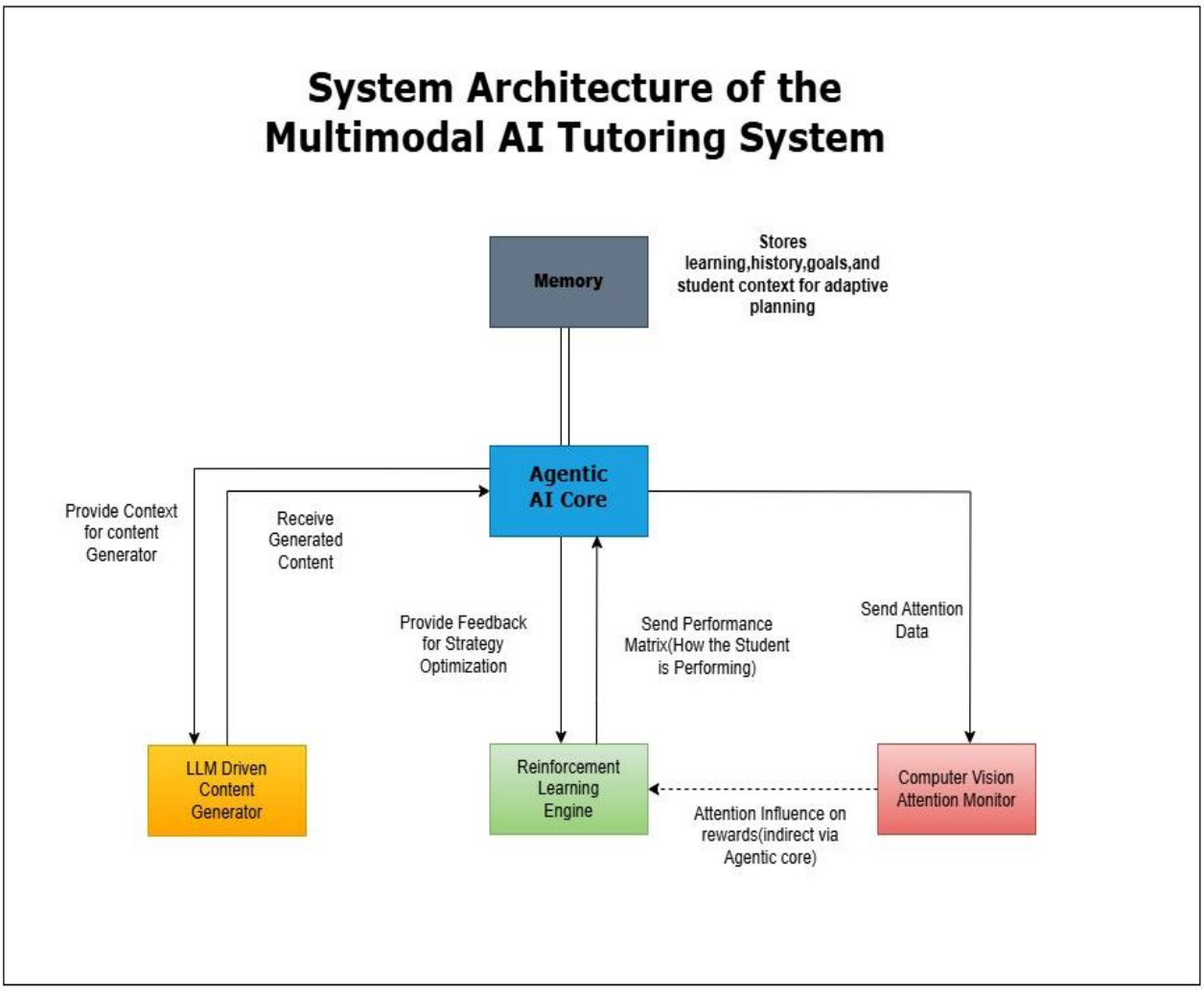
Agentic AI Core: This acts as the tutor's brain, simulating a goal-oriented teacher. It breaks down high-level learning objectives into multi-step instructional tactics using planning frameworks like Tree of Thought and ReAct. It continuously monitors student progress and adjusts the plan based on memory and predicted outcomes.

Reinforcement Learning (RL) Engine: By engaging with actual or simulated students, the RL component learns optimal teaching strategies. It uses a custom reward function based on engagement levels, performance, and learning retention. Over time, it evolves to tailor its decisions for each learner.

LLM-Driven Content Generator: A fine-tuned large language model generates quizzes, hints, summaries, and explanations personalized to the student's learning style, mood, and curriculum stage. It dynamically adapts tone, difficulty, and format to improve understanding and motivation.

Computer Vision Attention Monitor: A lightweight, privacy-aware vision model tracks facial cues like gaze and expressions to infer real-time engagement and affect. This data is fed into the agentic and RL modules to enhance responsiveness and empathy.

These components are orchestrated in a feedback loop, enabling the system to plan, act, observe, and adapt continuously. The result is a proactive tutor capable of delivering emotionally intelligent, goal-driven instruction across diverse learning contexts.


System Architecture of the Multimodal AI Tutoring System
Memory
Stores learning,history,goals,and student context for adaptive planning
Agentic AI Core
Provide Context for content Generator
Receive Generated Content
Provide Feedback for Strategy Optimization
Send Performance Matrix(How the Student is Performing)
Send Attention Data
LLM Driven Content Generator
Reinforcement Learning Engine
Computer Vision Attention Monitor
Attention Influence on rewards(indirect via Agentic core)

7.Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

This research requires complementary skills in AI planning, machine learning, natural language processing (NLP), and computer vision, along with a foundational understanding of educational psychology and instructional design.

The agentic AI core requires knowledge of goal-oriented planning algorithms, including ReAct, Tree of Thought, and goal decomposition techniques. Simulating long-term teaching tactics requires an understanding of agent architectures and memory-based reasoning.

An understanding of reinforcement learning fundamentals, including state-action-reward models, policy optimization, and custom reward design, is necessary for the reinforcement learning module. The RL agent will be trained using both simulated learner models and live student feedback, requiring experience in behavioral simulation and data-driven policy evaluation.

Proficiency in prompt engineering, fine-tuning, and alignment of large language models such as GPT is required for the LLM-driven content generator. Based on the learner profile, the system must produce curriculum-aligned and adaptable instructional content, which requires the use of NLP, educational text generation, and response evaluation skills.

The computer vision module uses emotion analysis and lightweight facial landmark detection to gauge involvement and attentiveness. Basic CV knowledge is necessary for this component, especially in the areas of gaze tracking, expression recognition, and real-time image processing using pre-trained models adapted for privacy-conscious usages.

Data requirements include:

- Simulated student behavior data for initial RL training.
- Sample tests and curriculum for creating content.
- Annotated facial engagement datasets (real or synthetic) for vision model fine-tuning.
- Human evaluation data (performance logs, feedback) for system validation.

Additionally, integration of these subsystems requires expertise in system architecture, feedback loop design, and real-time inference. Overall, the project spans both technical AI domains and pedagogical considerations, making it a rich, multidisciplinary challenge.

7.Objectives and Novelty

Main Objective To create a proactive, multimodal intelligent teaching system that uses computer vision, large language models, reinforcement learning, and automated planning to provide personalized, adaptive, and emotionally aware instruction.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
IT22915818 Dilshan B.A.T.	To design and implement an agentic AI core capable of decomposing learning goals into multi-step instructional strategies using planning frameworks.	• Develop a memory-based goal tracking mechanism. • Integrate planning techniques like Tree of Thought or ReAct for foresight-based instructional decision-making. • Simulate a goal-driven tutor capable of adjusting plans in real time. • Evaluate planning coherence, task completion efficiency, and learning path optimization.	Unlike reactive tutors, this component introduces proactive, human-like foresight through goal decomposition and adaptive planning. It allows the tutor to simulate strategic reasoning and continuously revise learning paths based on student progress and memory, addressing the gap of long-term instructional guidance in current ITS.
IT22356772 Kaushalya L.B.S.	To develop a content generation module with a fine-tuned large language model for personalized instruction.	• Employ a RAG framework that dynamically retrieves relevant educational material. • Use prompt tuning for adjusting tone, difficulty, and explanation style.	This component changes generic LLM outputs into focused, curriculum based instruction. It addresses the limitations of reactive chatbots by actively directing content generation. It does this using learning goals and student mood.

		• Create quizzes, summaries, and hints that fit the learner's context. • Assess the teaching quality and learning outcomes of the generated content.	The result is personalized and context aware tutoring.
IT22916594 Gunawardena D.T.	To develop a privacy conscious, multimodal engagement monitoring module that combines lightweight Computer Vision (CV) techniques with Vision-Language Models (VLMs) to interpret real-time learner affect, attention, and cognitive state for adaptive tutoring.	• Design a lightweight CV pipeline to extract attention related cues like gaze direction, facial expressions, and posture using tools such as MediaPipe or OpenPose. • Integrate Vision-Language Models, like CLIP or LLaVA, to process visual input along with textual context, such as current lesson material or student queries. • Develop a multimodal engagement interpretation engine that combines CV outputs with VLM contextual outputs to understand student states like confusion, boredom, or interest. • Set up real-time communication links between this module and the Agentic AI Core and RL	Unlike traditional Intelligent Tutoring Systems (ITS), this component integrates silent, non-intrusive visual feedback directly into the tutor's decision-making process. It improves adaptivity by detecting early signs of disengagement or confusion. This enables the system to respond both emotionally and in context. This bridges the affective gap that often exists in digital learning environments. It allows for more human-like and responsive tutoring.

		Engine to influence teaching strategies. • Implement privacy-preserving methods, such as on-device inference, data anonymization, and optional user controls for ethical compliance. • Evaluate the accuracy and effectiveness of engagement interpretation using annotated datasets. Test its impact on learning outcomes and tutor responsiveness.	
IT22334060 Kuruppuarachchi A.C.	To create a reinforcement learning engine that improves teaching strategies based on real-time engagement and performance feedback.	• Create a custom reward function that includes attention, performance, and retention metrics. • Train the RL model with simulated student data, then refine it with real user feedback. • Incorporate RL outputs into the tutor's decision loop to enable adaptive instruction. • Compare strategies driven by RL with those based on rules.	While traditional tutors depend on fixed methods, this RL engine allows the tutor to keep learning from interactions. It improves teaching decisions over time. It fills the gap of real-time flexibility and long-term improvement in instruction.

7. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
IT22915818 Dilshan B.A.T.	The Agentic AI Core uses Data Science principles to model, simulate, and improve decision-making in educational settings. It applies planning algorithms like Tree of Thought and ReAct to break down learning goals into data-driven teaching plans. The system relies on past student data, performance metrics, and contextual memory to adjust its approaches in real time. This includes creating models to track goals, finding patterns in student behavior, and confirming results through performance analysis. The component shows key Data Science skills such as predictive modeling, smart planning, and system evaluation through experimentation. It fits well with the specialization's focus on analytical thinking, AI integration, and practical machine learning.
IT22356772 Kaushalya L.B.S.	The content generator uses NLP and data-driven personalization techniques, both central to Data Science. It involves prompt engineering, fine-tuning transformer-based language models, and adapting content based on structured data inputs such as curriculum, student profile, and emotional state. Data Science expertise helps create relevant and pedagogically sound outputs, assess textual quality, and ensure effectiveness through empirical testing. This component showcases natural language generation, response evaluation, and A/B testing, which are key areas in Data Science. It connects AI language modeling with educational personalization through ongoing data-driven adjustments.
IT22916594 Gunawardena D.T.	This component focuses on using computer vision to measure real-time engagement. It employs data science methods like image data preprocessing, classification, and real-time signal analysis. By extracting features such as gaze direction, head pose, and facial expression, it represents attention and emotions as measurable variables. These metrics are then linked to performance data to improve tutor adaptivity. The process includes supervised learning, data labeling, and model evaluation, which reflect common data science workflows. This module shows how multimodal data, both visual and behavioral, can be analyzed and understood for smart decision support in learning environments.
IT22334060 Kuruppuarachchi A.C.	This component uses key Data Science principles to improve decision-making in real-time tutoring. It applies supervised and reinforcement learning methods to track student engagement and performance as feedback signals. The reinforcement learning agent trains on both simulated and real interaction data, using performance analysis to develop flexible teaching strategies. Creating reward functions, training models, and assessing long-term results require data preparation, behavior modeling, and ongoing testing; these are all essential to Data Science. This component shows how machine learning

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	works in practice, along with dynamic system improvement and using feedback loops to enhance intelligent systems in education.

9. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Mr	Jeewaka	Perera	
Co-Supervisor	Dr.	Kalpani	Manathunga	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes		No	

b) Does the proposed topic exhibit novelty?

Yes		No	

c) Do you believe they have the capability to successfully execute the proposed project?

Yes		No	

d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes		No	

e) Supervisor's Evaluation and Recommendation for the Research topic:

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Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

*__Important__:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.